CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Transpiration pull is chiefly carried out by: 1

(a) phloem **(b)** xylem **(c)** cork **(d)** epidermis

2. WBCs help in: 1

(a) digestion **(b)** clotting **(c)** carrying oxygen **(d)** fighting germs

3. The axon of a neuron: 1

(a) carries impulses away from the cell body **(b)** stores food

(c) contracts **(d)** receives impulses

4. Cells in the root cap are: 1

(a) photosynthetic **(b)** conducting

(c) protective **(d)** dividing rapidly only

5. Aquatic plants have tissue with air spaces called: 1

(a) cork **(b)** collenchyma **(c)** sclerenchyma **(d)** aerenchyma

6. Platelets help in: 1

(a) defence **(b)** carrying oxygen

(c) clotting of blood **(d)** digestion

7. The dendrites of a neuron: 1

(a) receive impulses **(b)** store energy **(c)** make proteins **(d)** send impulses

8. A potato tuber stores starch in: 1

(a) parenchyma **(b)** xylem **(c)** sclerenchyma **(d)** collenchyma

9. A guard cell differs from other epidermal cells in having: 1

(a) chloroplasts **(b)** a nucleus **(c)** a cell wall **(d)** a vacuole

10. The muscle that does not show fatigue is the: 1

(a) skeletal muscle **(b)** smooth muscle only

(c) cardiac muscle **(d)** all muscles equally

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do guard cells have chloroplasts? 2

12. State the role of the companion cells. 2

13. Why are the red blood cells without a nucleus? 2

14. Why do aquatic plants have well-developed aerenchyma? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the role of the epithelial tissue in skin, intestine and lungs. 3

16. How does the structure of the root hair suit its function? 3

17. State the difference between a tissue and an organ and give examples. 3

OR

Describe the structure of bone and cartilage and their functions. 3

18. Explain why cork cells are dead and impermeable. 3

OR

Explain the use of stains to see tissues under a microscope. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the process of healing of a bone fracture and a skin wound and the tissues involved. 5

OR

Explain how the structure of the leaf tissues (epidermis, palisade, spongy, vascular, stomata) is adapted for photosynthesis and gas exchange. 5

20. Explain, giving reasons, why the nervous tissue is called the control and coordination tissue and describe a neuron and the passage of a signal in a reflex action. 5

OR

Discuss the application of tissue culture and stem cells in agriculture and medicine. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND